HOLY CROSS COLLEGE

Life to the Full

12 PHYSICS

ELECTRICITY & MAGNETISM

ELECTRIC FIELDS

A charged object has an electric field associated with it, and this influences other charged objects placed in it.

The electric field strength is defined as the force per unit charge.

i.e. $$\boxed{E = \frac{F}{q}}$$

where E = the electric field intensity (NC^{-1})
F = the force exerted on the charged object in the electric field (N)
q = the size of the charge on the object in the electric field (C)

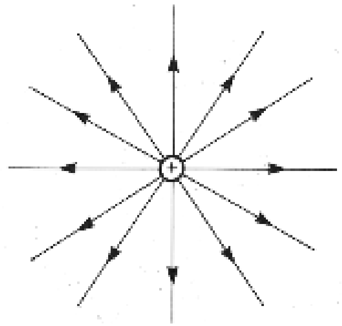
Electric field has both size (strength) and direction, hence it is a vector.

The direction of an electric field is taken as the direction a positive test charge, when placed in the field, would move.

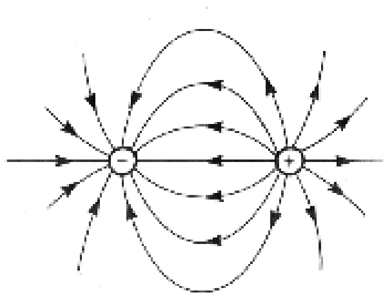
i.e. away from a positively charged object.

An electric field is represented by lines of force or flux lines.

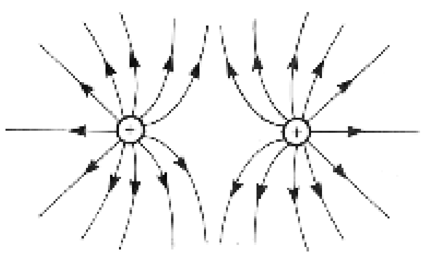
The electric field around various point and plate charges are as follows.



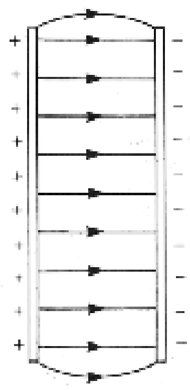
Isolated positive charge



Opposite point charges



Similar point charges



Parallel plates

Rules For Drawings

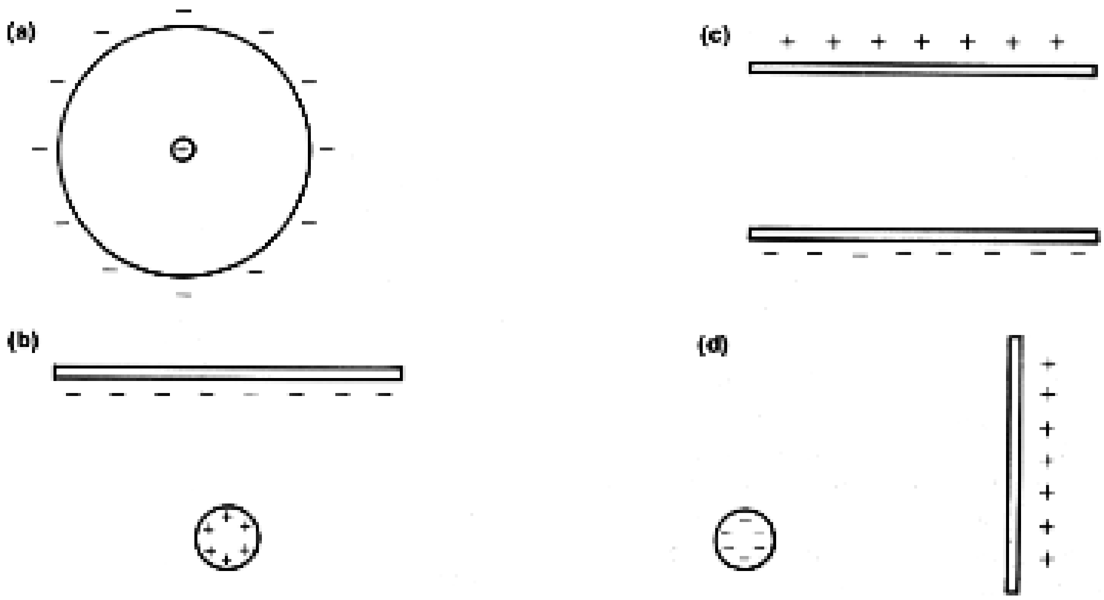
1. Electric field lines never touch nor cross each other.

2. They enter and leave surfaces at right angles.

3. The intensity of the electric field is represented by the density or "bunching" of the field lines.

Examples

Draw the electric fields associated with the following.



ELECTRIC POTENTIAL

The Earth is defined as having zero potential.

Charged bodies gain electrical potential energy when they are moved in an electric field.

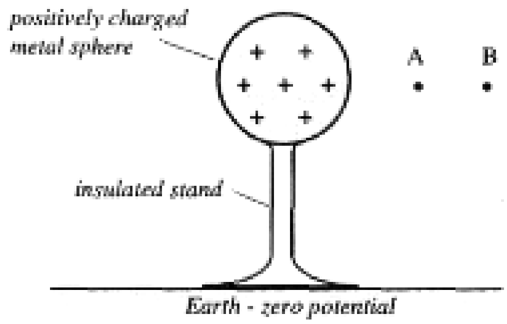
Consider a positively charged body that cannot move and is insulated from the Earth.

If a positive charge is brought to point B, work has to be done to overcome the repulsion.

This work is stored as electric potential energy.

If it is moved to point A, it has a higher electrical potential energy.

Points A and B have different potentials - there is a **potential difference**.



Potential difference is therefore defined as the work done per unit charge in moving the charge through an electric field.

i.e. $$\boxed{\mathbf{V = \frac{W}{q}}}$$

where
V = the difference in potential between two points in an electric field (V - volts)
W = work done in moving the charge (J)
q = the amount of charge moved in the electric field (C)

When a charge is placed in an electric field, it experiences a force and moves. It moves from one potential to another, thus requiring some work to be done.

If work is done in moving the charge, it must change E_k.

i.e. $$W = \Delta E_k$$

$$\Rightarrow \quad Vq = {}^1/_2\, mv^2 - {}^1/_2 mu^2$$

Since the charge starts from rest, ${}^1/_2\, mu^2 = 0$.

$$\therefore \quad \boxed{W = Vq = {}^1/_2 mv^2}$$

This Physics principle of accelerating charges is best considered in the electron gun inside an old-style TV (cathode ray tube).

Example
The electrons inside the "red" electron gun of a standard colour TV are accelerated by a potential difference of 1.32×10^3 V. Determine how fast these electrons leave the gun to move down the picture tube.

ELECTROMOTIVE FORCE (EMF)

In order to move charges, a source of electrical potential energy must do work on them.

Such sources include torch cells, solar cells, photovoltaic cells, generators, piezoelectric crystals, and thermocouple junctions.

These devices are referred to as sources of ***electromotive force*** or ***EMF***. This is a misnomer as the term refers to the energy given to the charges rather than any force that is applied to them.

EMF is defined as the amount of work done per unit charge in moving the charges through the source.

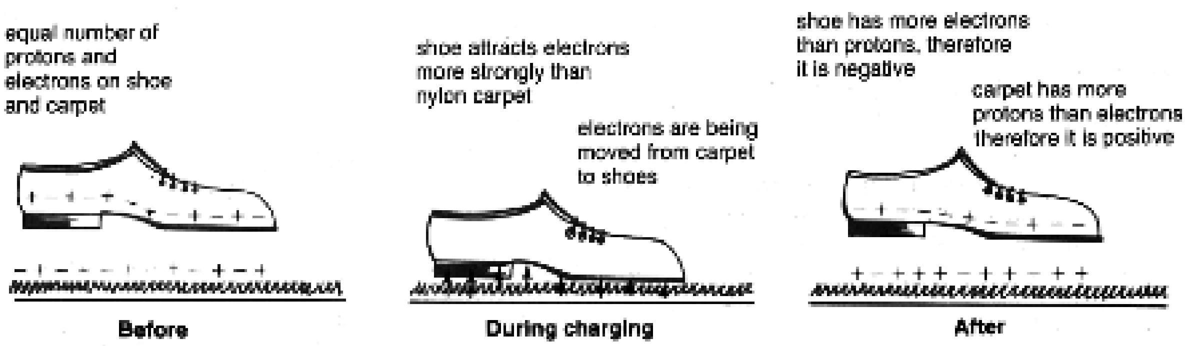
STATIC ELECTRICITY

The outer electrons of atoms are very easily removed as they are held relatively weakly by the attraction of the protons in the nucleus. This is easily achieved by rubbing materials together.

When two materials are rubbed together, electrons are transferred between them. One material will gain electrons (and gains a nett negative charge due to having more electrons than protons present), and the other will lose electrons (and gain a nett positive charge).

This build up of charge is called ***static electricity***.

e.g. Consider the build up of charge when walking on certain carpets.

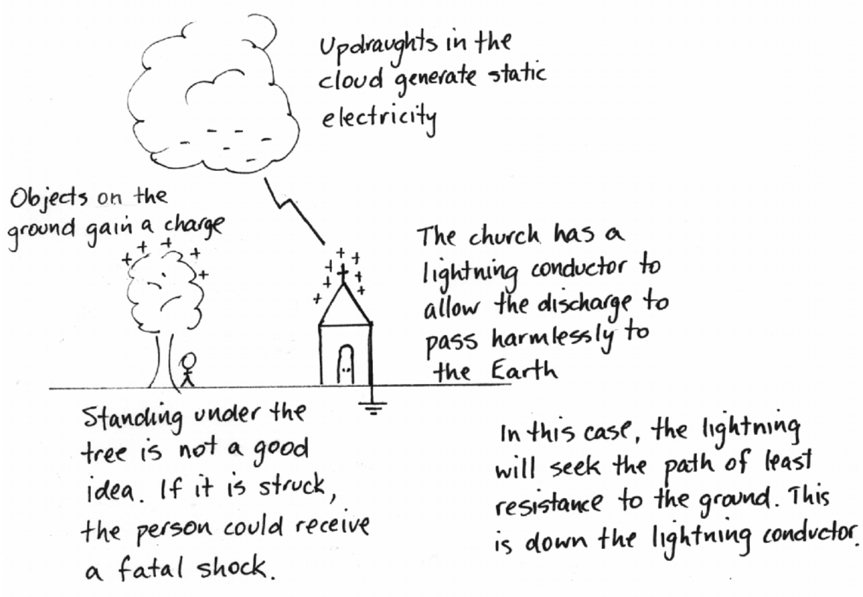


Law Of Electrostatics

Like charges repel, unlike charges attract.

If the charge build up on two objects is great enough, and these charges are opposite in sign, then a spark may jump between the two.

It is this effect that is responsible for lightning discharging between clouds and the Earth.



Fundamental Charge

The elementary charge (e) is given as that charge on a proton or electron.

i.e. $e = 1.60 \times 10^{-19}$ C (Where C represents the units - Coulombs)

FORCE BETWEEN CHARGES

Coulomb's Law is used to describe the force between charged objects.

i.e.
$$\boxed{F = \frac{1}{4\pi\varepsilon_o}\frac{q_1 q_2}{d^2}}$$

where

F = the force of attraction or repulsion (N)

q_1, q_2 = the charges on the objects (C)

d = the distance between the two objects (m)

ε_o = the dielectric constant (permittivity) for the medium ($C^2N^{-1}m^{-2}$)

Note

The permittivity of air (ε_{air}) is almost the same as that of a vacuum (ε_o).

Therefore, in calculations use:

$$\varepsilon_{air} = \varepsilon_o = 8.85 \times 10^{-12}\ C^2N^{-1}m^{-2}$$

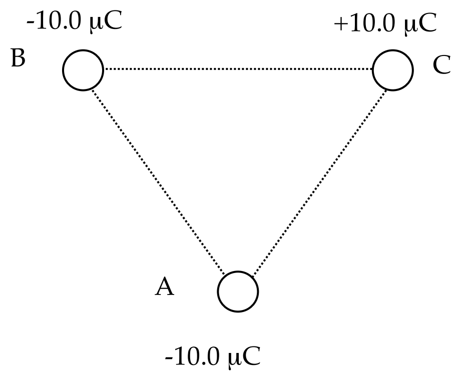
Remember

If the charges are similar in sign, the force is repulsive.

If the charges are opposite in sign, the force is attractive.

Example
Three point charges are arranged in air as shown below. They are at the vertices of an equilateral triangle of side 10.0 cm.

Calculate the force on the charge at A due to the presence of the other charges.
(Assume that the charges are fixed and can't move relative to each other.)



PROPERTIES OF MAGNETS

A freely-suspended piece of magnetic material will always come to rest pointing along a north-south line with respect to the Earth's magnetic field. This is called the ***directional property of magnets.***

The end that points to the north pole is called the ***north-seeking pole***, leading to it being called the ***north pole of the magnet***.

Iron filings scattered over a magnet will concentrate near the ends of the magnets.
i.e. ***The magnetic force is strongest at the poles***.

The poles of magnets will interact if brought near one another.
i.e. ***Like poles repel, unlike poles attract***.

MAGNETIC MATERIALS

Magnetic materials are those that can be magnetised.

e.g. Fe, Co, Ni and certain alloys of these elements show strong magnetic properties.

Magnetic materials can be classified as follows.

1. **Ferromagnetic** - strongly attracted.

   e.g. Fe, Ni, Co, alnico (alloy of Al, Ni, Co, Fe), permalloy (alloy of Fe, Ni)

2. **Paramagnetic** - weakly attracted.

   e.g. Mn, Al, liquid O_2, Pt

3. **Diamagnetic** - weakly repelled.

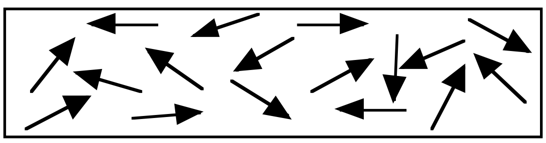
   e.g. Cu, Au, Bi

MOLECULAR THEORY OF FERROMAGNETISM

Ferromagnetic materials have their atoms grouped in isolated fragments called ***domains***.

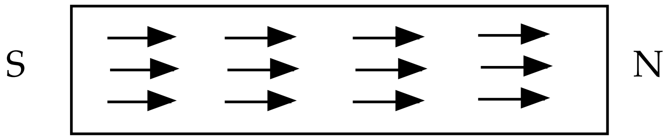
Within each of these domains, the atoms are aligned in such a way that their magnetic fields (***due to their moving and spinning electrons - see notes on Magnetic Fields of Current-Carrying Wires.***) combine, so that each domain can be thought of as a tiny magnet.

In an unmagnetised material, the domains are arranged randomly.



UNMAGNETISED IRON

When a magnetic field is applied to the material, the domains align so that they point in the same direction as the field. One end becomes the north pole, the other the south pole.



MAGNETISED IRON

Physics in action — Magnetic domains

Ferromagnetic substances include iron (and some of its compounds and alloys), nickel, cobalt and gadolinium. Although all atoms have spinning electrons, only a small number of materials are ferromagnetic, and not all pieces of iron are magnets. It turns out that while almost all of the magnetic dipoles do line up with their neighbours, they do it in large groups of atoms, not right throughout the whole piece of metal. The groups contain huge numbers of atoms (around a billion billion, or 10^{18}), but are still rather small in size (of the order of micrometres, 10^{-6} m). These groups of atoms are called *domains*. Normally, the directions of their dipoles are random and so their fields all cancel out.

Each domain has a very strong (but tiny) magnetic field, but if all the domains are oriented differently, the overall effects will cancel out. When the iron is put into a magnetic field, two things can happen: the domains whose magnetism is already pointing in the direction of the field may grow by (in a manner of speaking) taking over nearby groups of atoms, or whole domains may switch their orientation so that they are more in line with the field. The extent to which these things happen determines how strong the magnet will become.

When the external field is removed, the domains may flip back to their random directions (e.g. in soft iron) or they might retain their new directions (e.g. in hard iron). Permanent magnets need to be made out of iron that is as 'hard' as possible; that is, iron in which the domains are very resistant to changes in their orientation once the iron is cool. They are made by heating the iron in a strong field. The heat enables the domains to change more easily.

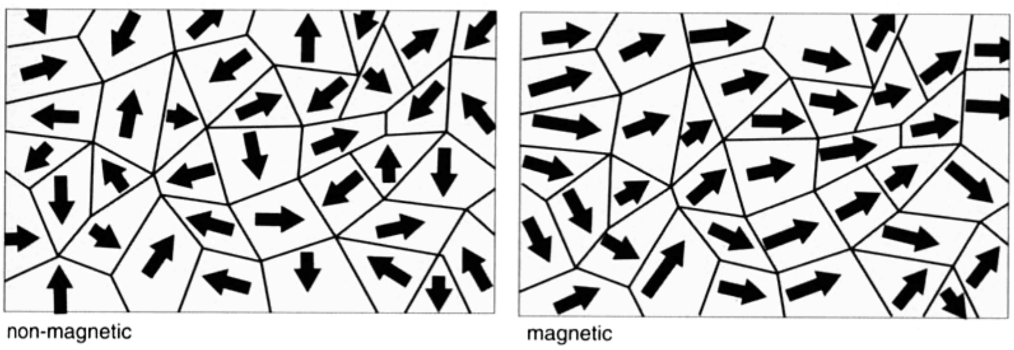


Figure 3.18
When placed in an external field, the domains that are oriented in the right direction grow and the others either shrink or change direction.

LINES OF MAGNETIC FLUX

When iron filings are sprinkled onto a sheet of paper with a bar magnet underneath, they align themselves with the magnetic field present.

Any region in which a magnetic material experiences a ***force*** due to its proximity to a magnet is said to contain a magnetic field.

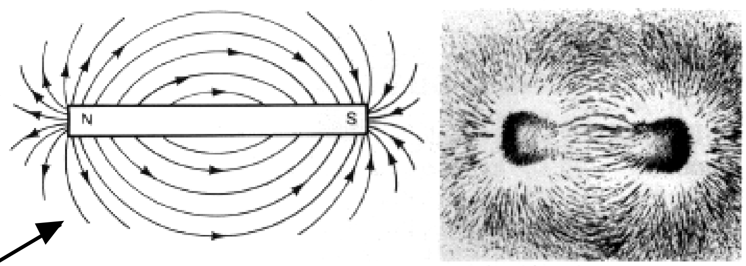
The direction of the field at any point is taken as the direction of a force on the north pole of a magnet placed in the field at that point.

Magnetic fields are shown using lines of magnetic flux. These are lines whose shape is such that the tangent at any point gives the direction of the magnetic field at that point.

Arrowheads on the flux lines show the direction of the force acting on the north pole of a small test compass needle placed at that point.

i.e. ***The flux lines are always out of north and into south***.

The field lines are closer together near the poles, indicating a stronger magnetic field.



The field lines never touch each other.

The direction of the magnetic field is always out of north and into south.

To fully describe a magnetic field at a point, both a magnitude (or strength) and a direction must be given.

i.e. It is a vector quantity.

This vector is called the ***magnetic flux density (B)***. It can also be referred to as ***magnetic intensity*** or ***magnetic induction***.

The magnitude of the magnetic flux density is given as the number of flux lines passing through a unit area.

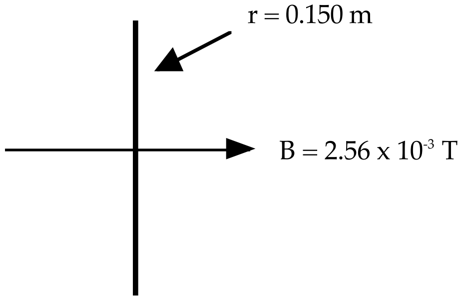
In a uniform magnetic field:

$$\boxed{\phi = BA}$$

where
ϕ = magnetic flux (Webers - Wb)
B = magnetic flux density (Wbm^{-2} or Tesla - T)
A = area through which the flux acts (m^2)

Example 1

A circular coil of diameter 30.0 cm is arranged to be at right angles to a magnetic field of strength 2.56×10^{-3} T. Calculate the total amount of flux passing through it.

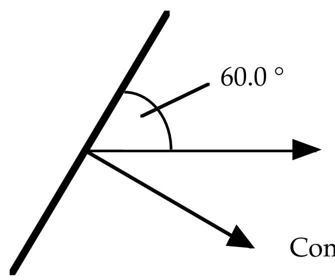


$$\begin{aligned}\Phi &= BA \\ &= (2.56 \times 10^{-3})\,\pi\,(0.150)^2 \\ &= 1.810 \times 10^{-4}\text{ Wb}\end{aligned}$$

$\therefore$ Flux threading the coil = 1.81×10^{-4} Wb

Example 2

If the coil in example 1 is changed to be at 60.0 ° to the field, what is the new value for the flux threading the coil?



To calculate the flux threading the coil, find the **component** of B which is at right angles to the coil.

$$\begin{aligned}\Phi &= BA \\ &= (2.56 \times 10^{-3} \cos 30.0\,°)\,\pi\,(0.150)^2 \\ &= 1.567 \times 10^{-4}\text{ Wb}\end{aligned}$$

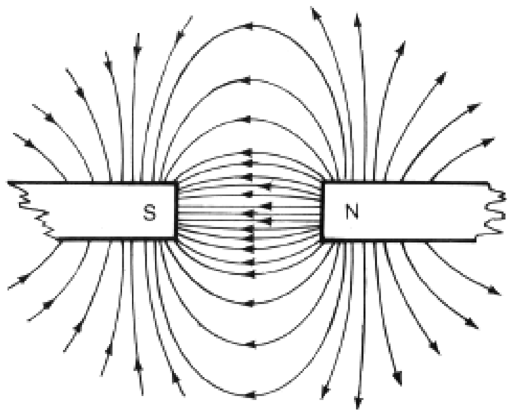
$\therefore$ Flux threading the coil = 1.57×10^{-4} Wb

INTERACTION BETWEEN MAGNETIC FIELDS

1. **Unlike Poles**

The fields combine to produce a new magnetic field in which the lines of magnetic force become denser.

∴ The magnets attract one another.

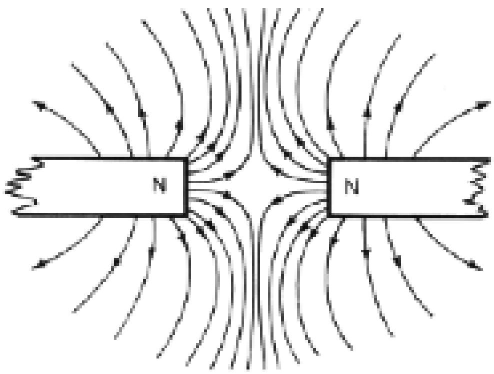


Notice that the field lines never touch.

2. **Like Poles**

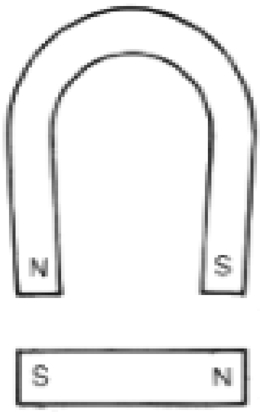
The magnetic fields are operating in opposite directions, resulting in a new magnetic field in which the lines of magnetic force become less dense.

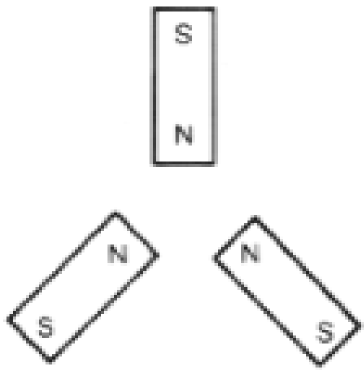
∴ The magnets repel one another.



Examples

1. Draw the magnetic fields that exist between the magnets in the following situations.





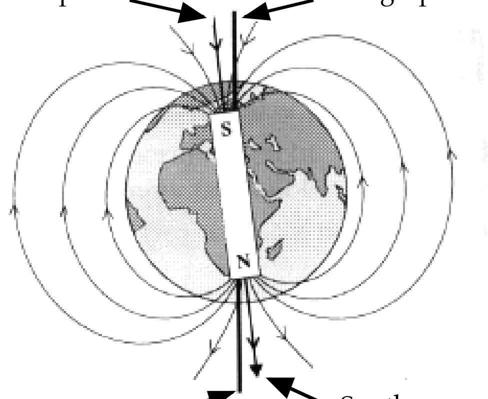
2. Draw the magnetic fields associated with the following.

   (a) A circular magnet.

   (b) A flat plastic fridge magnet.

EARTH'S MAGNETIC FIELD

The Earth is surrounded by a magnetic field known as the ***magnetosphere***. It is thought that electric currents generated within the core of the Earth cause the magnetic field to arise.



Note how the magnetic north pole acts as if it is the south pole of a bar magnet.

As stated previously, the directional property of a magnetic material arises because the north-seeking pole is attracted to the magnetic north pole of the Earth.

Is this wrong, considering that like poles repel? No!

The north-seeking pole is attracted to a south pole.

$\therefore$ The north magnetic pole of the Earth acts as if it is the ***south pole of a bar magnet***.

Consequently, when drawing the Earth's magnetic field, the field lines ***come out of the south magnetic pole and enter into the north magnetic pole***.

Notice also that the geographic poles and the magnetic poles do not coincide. In fact the magnetic poles continue to "wander" each year, changing their positions relative to the true north and south poles.

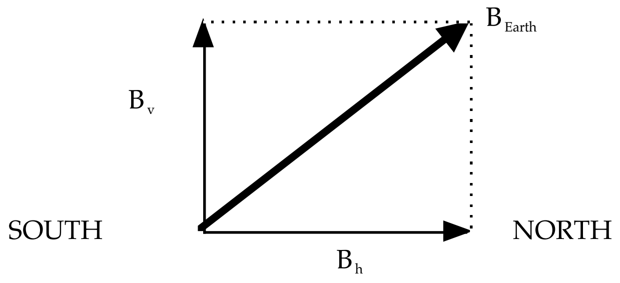
Angle Of Declination: the angle between true north and magnetic north.

This angle depends on the point on the Earth's surface, but all maps will carry "corrections" to allow navigators to find true north. Maps need to have these "corrections" updated each year as the magnetic poles move.

Any freely-suspended magnetic needle can also "dip" at an angle to the horizontal as it follows the Earth's magnetic field line.

Angle Of Dip: angle between the horizontal and the line of the magnetic field at that point.

Consequently, the ***Earth's magnetic field has a vertical and horizontal component*** at all points except at the magnetic poles.

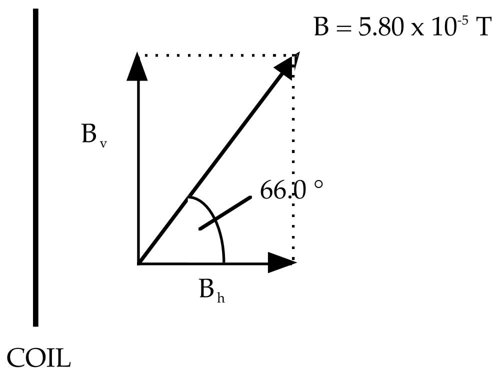


LOOKING WEST

In the Southern Hemisphere, the Earth's magnetic field is **angled upwards** to the horizontal and runs from **south to north**.

Example

A circular coil of radius 25.0 cm is arranged vertically in an east-west direction in Perth. Calculate the amount of flux that threads the coil, given the magnetic field intensity at Perth is 5.80×10^{-5} T at 66.0° dip.



LOOKING WEST

The horizontal component of B_{earth} is threading the coil.

$$\begin{aligned}\Phi &= BA \\ &= (5.80 \times 10^{-5} \cos 66.0°)\ \pi\ (0.250)^2 \\ &= 4.632 \times 10^{-6}\ \text{Wb}\end{aligned}$$

$\therefore$ Flux threading the coil $= 4.63 \times 10^{-6}$ Wb

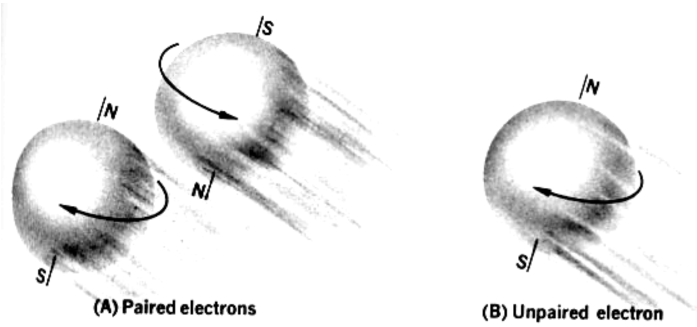
MAGNETIC FIELD AROUND A CURRENT-CARRYING CONDUCTOR

All moving charges produce a magnetic field, whether within a conductor or moving through space as in an atom.

i.e. ***Magnetism is created by moving charges.***

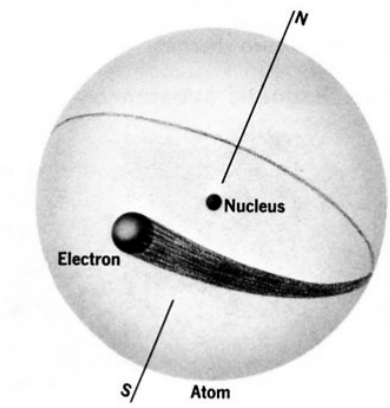
e.g. Spinning electrons in an atom produce a field. If electrons are paired, they spin in opposite directions and their fields cancel.

Ferromagnetic atoms such as iron have unpaired electrons that produce a magnetic field for the atom.



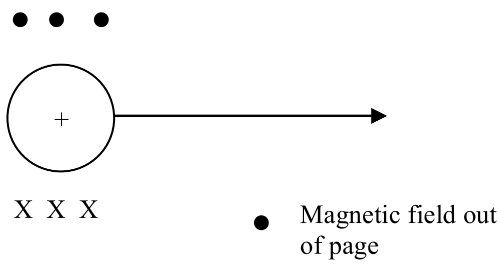
19-2 Magnetism in matter stems basically from the spin of electrons.

Electrons orbiting in an atom produce a field.



19-1 Revolving electrons impart a magnetic property to the atom.

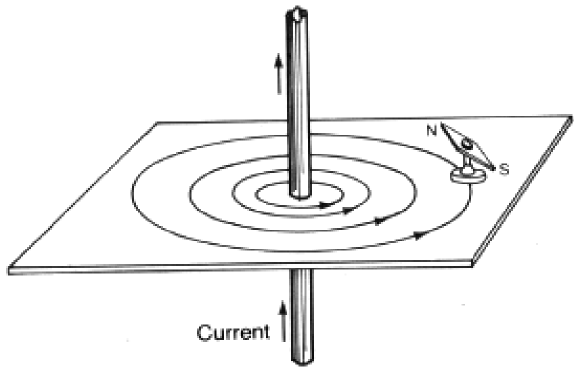
A moving positive charge produces a field.



Note

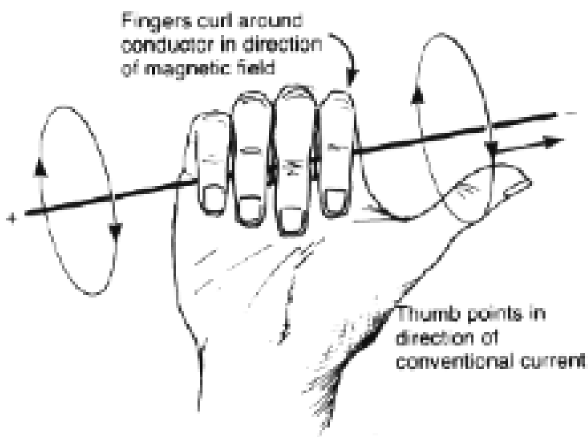
The direction of the field for an isolated charge is given by the Right Hand Curl Rule - see notes ahead!

The magnetic field produced by a current-carrying wire is a series of concentric circles centred on the wire. It is produced by the moving charges inside.



The current flowing in the wire is assumed to be conventional current - the flow of positive charges.

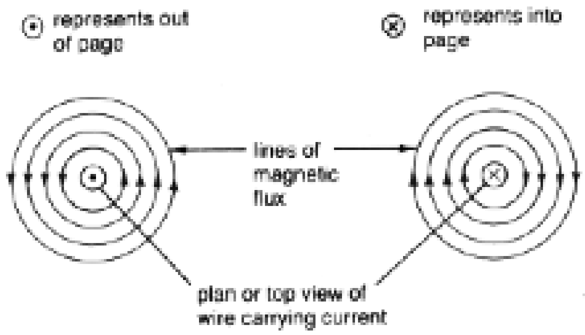
The direction of the magnetic field is given by the ***right hand grasp rule***.



Note

Conventional Current: the flow of ***positive charge*** from the ***positive terminal to the negative terminal*** of the source of the current.

It is easier to represent the magnetic field due to a current by a two-dimensional diagram.



Note
The convention for the direction of the conventional current is as follows.

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Make sure that you put one of these labels on the bottom of every diagram you draw so that there is no confusion for the person marking your test or exam paper.

Calculating Magnetic Field Strength

Like most fields, the closer an object is to a current-carrying wire, the stronger the magnetic field. It is also observed that the amount of current in a wire affects the magnetic field strength.

The strength of the magnetic field (B) around a wire is directly proportional to the current (I) in a long, straight conductor and inversely proportional to the distance from the conductor (r). That is:

$$\boxed{B = \frac{\mu_o I}{2\pi r}}$$

where B = strength of the magnetic field in tesla (T)
I = current in the conductor in ampere (A)
r = distance from the conductor in metres (m)
μ_0 = the permeability of free space, $\mu_0 = 4\pi \times 10^{-7}$ TmA^{-1}

B is a vector quantity, the direction being that of the magnetic field around the conductor. It is a measure of the magnetic flux density.

Example

A current of 5.00 A is running through a long straight wire. Determine the magnitude of the strength of the magnetic field at a point 10.0 cm from the wire.

B = ?

I = 5.00 A

r = 10.0 cm = 0.100 m

$$B = \frac{\mu_o I}{2\pi r}$$

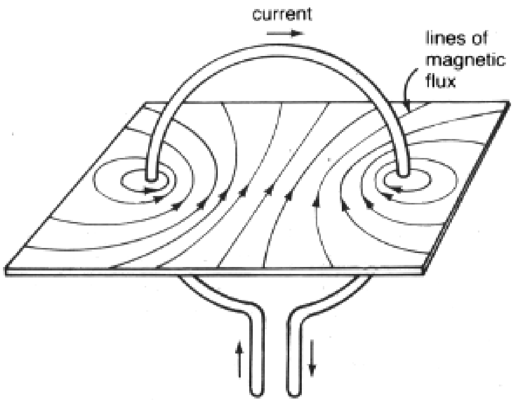
$$= \frac{(4\pi \text{ x } 10^{-7})(5.00)}{2\pi(0.100)}$$

$$= 1.00 \text{ x } 10^{-5} \text{ T}$$

$$\therefore \quad \underline{B = 1.00 \text{ x } 10^{-5} \text{ T}}$$

MAGNETIC FIELD AROUND A CURRENT LOOP AND A SOLENOID

Single Loop

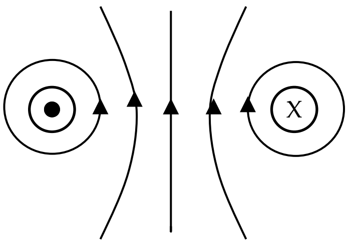


The magnetic field is shown at left.

The strength of the field is increased in the middle of the loop.

This is due to all the field vectors moving in the same direction and hence add together.

It is much easier to show this field in a two-dimensional drawing.



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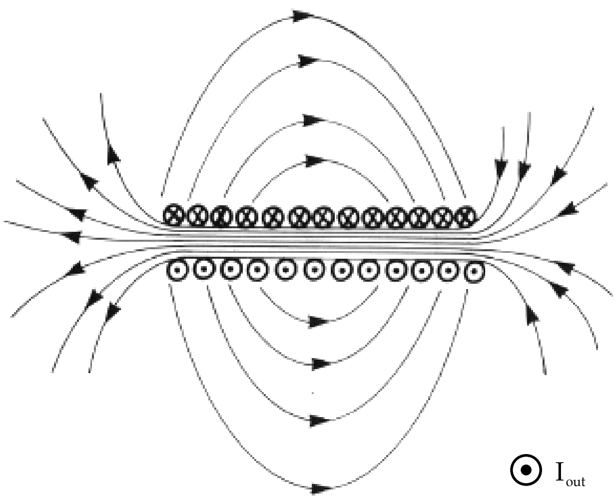
The strength of the field can also be made stronger by ***increasing the number of loops*** to make a flat coil.

This leads to the formation of a ***solenoid***.

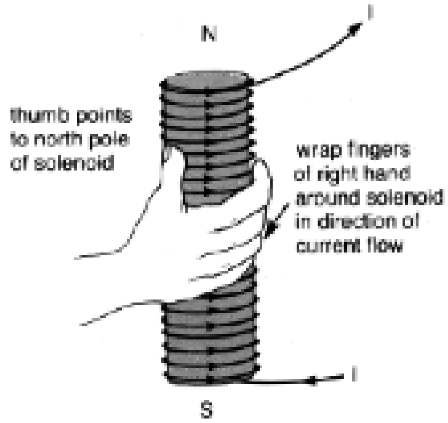
Solenoid

This contains many loops arranged in a tight cylindrical coil.

It has a uniform field inside and the external field is similar to a bar magnet.



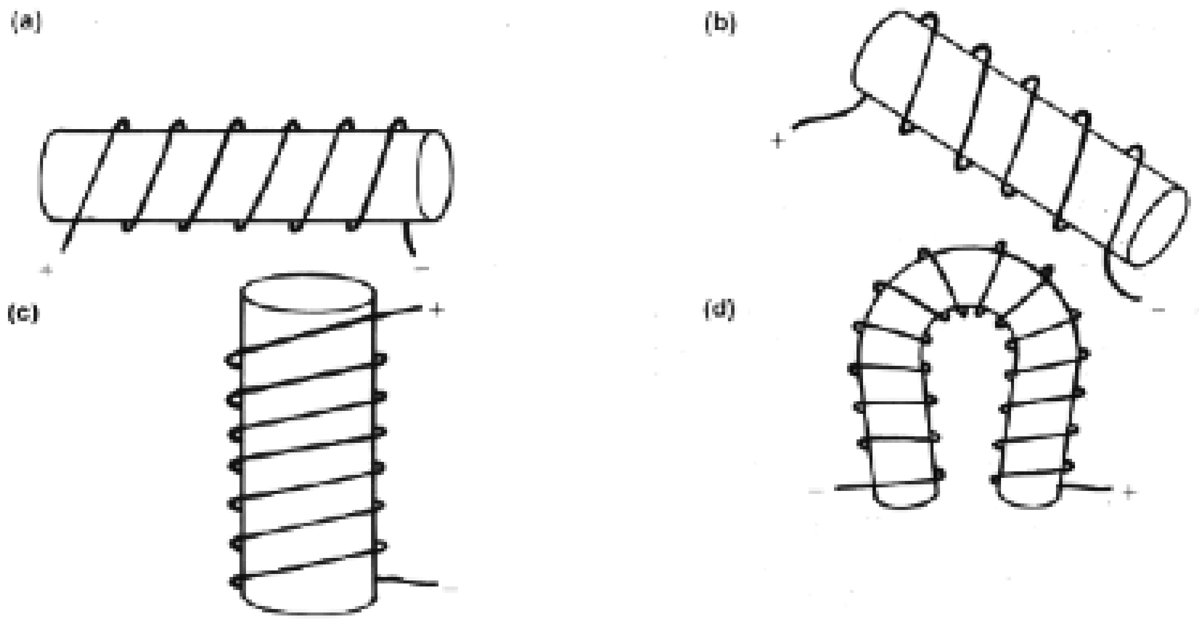
Right Hand Curl Rule: Curl the fingers of the right hand around the solenoid in the direction of the conventional current flow and the thumb will point towards the north pole.



Note

Care must be taken to clearly identify the direction of the conventional current flow.

Examples
On the following diagrams, show the N and S poles.



FORCE BETWEEN PARALLEL CONDUCTORS

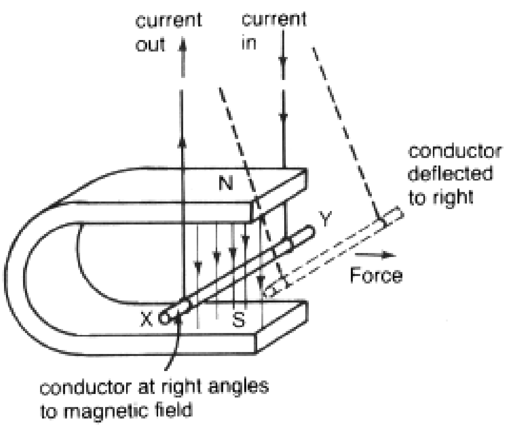
The French physicist Ampere noticed a strange phenomenon when two current-carrying wires were placed parallel and close to each other.

(a) Current in the ***same direction*** in the wires - ***attraction occurred***.

(b) Current in the ***opposite directions*** - ***repulsion occurred***.

The reason becomes obvious if one remembers that each wire generates a magnetic field and this can interact with any other magnetic field.

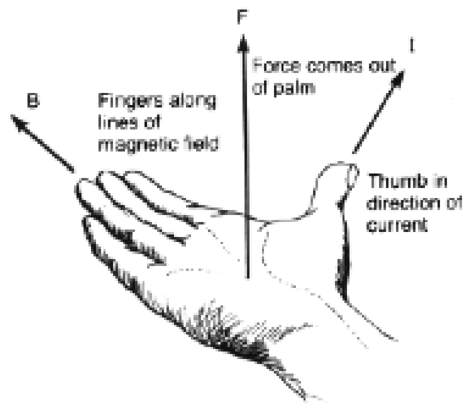
Consider the following experimental arrangement.



The conductor moves outwards from the magnetic field as its magnetic field interacts with that of the permanent magnet.

To determine in which direction the conductor will move, the ***right hand palm*** rule is used.

Right Hand Palm Rule: Fingers point in the direction of B, thumb points in the direction of conventional current flow.
Palm points in the direction of the force exerted on the conductor.

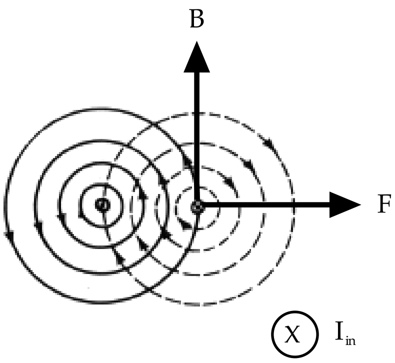


Note

More detail will be given on this in the next section of work.

We can apply this understanding to parallel conductors.

i.e. Currents in ***opposite directions***.

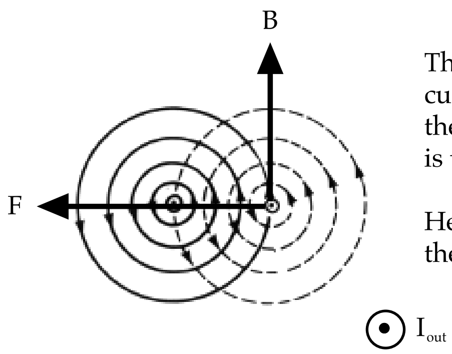


The conductor on the right is in the magnetic field of the other conductor. Consequently, if I is into the page and B due to the other conductor is up the page, the force experienced is away from the other conductor.

The same applies to the other conductor.

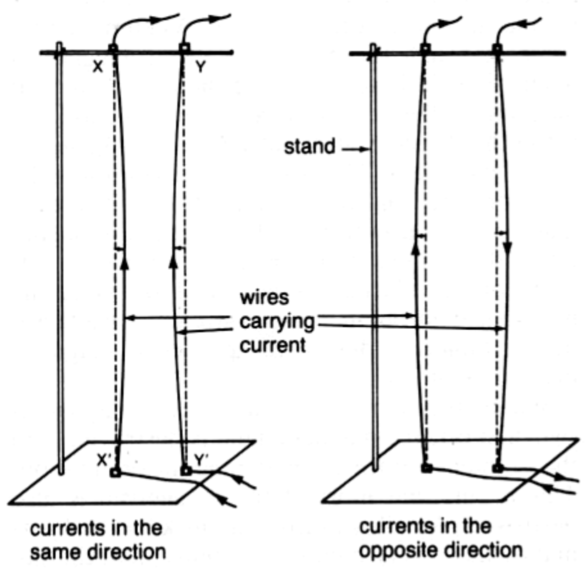
Hence the conductors **repel** each other.

Currents in the ***same direction***.



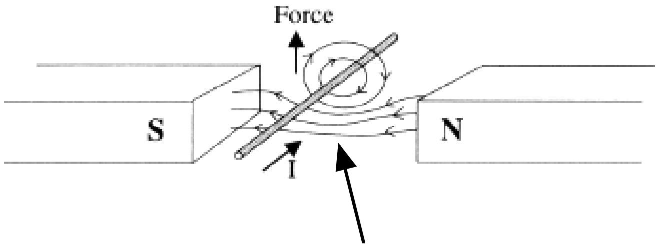
The conductor on the right has the current coming out of the page. It is in the field of the other conductor which is up the page.

Hence the force experienced is **towards** the other conductor.



FORCE ON A CONDUCTOR IN A MAGNETIC FIELD

As seen before, a current-carrying conductor in a magnetic field produces its own field that interacts with the permanent field.



The magnetic fields add together, creating a **more dense** and hence **stronger** field. To minimise the "stress" in the field, the conductor is pushed upwards.
i.e. From area of high density to low density.

Remember that the direction of the force on the conductor is given by the ***right hand palm rule***.

The force exerted on the conductor is given by:

$$\boxed{F = IlB}$$

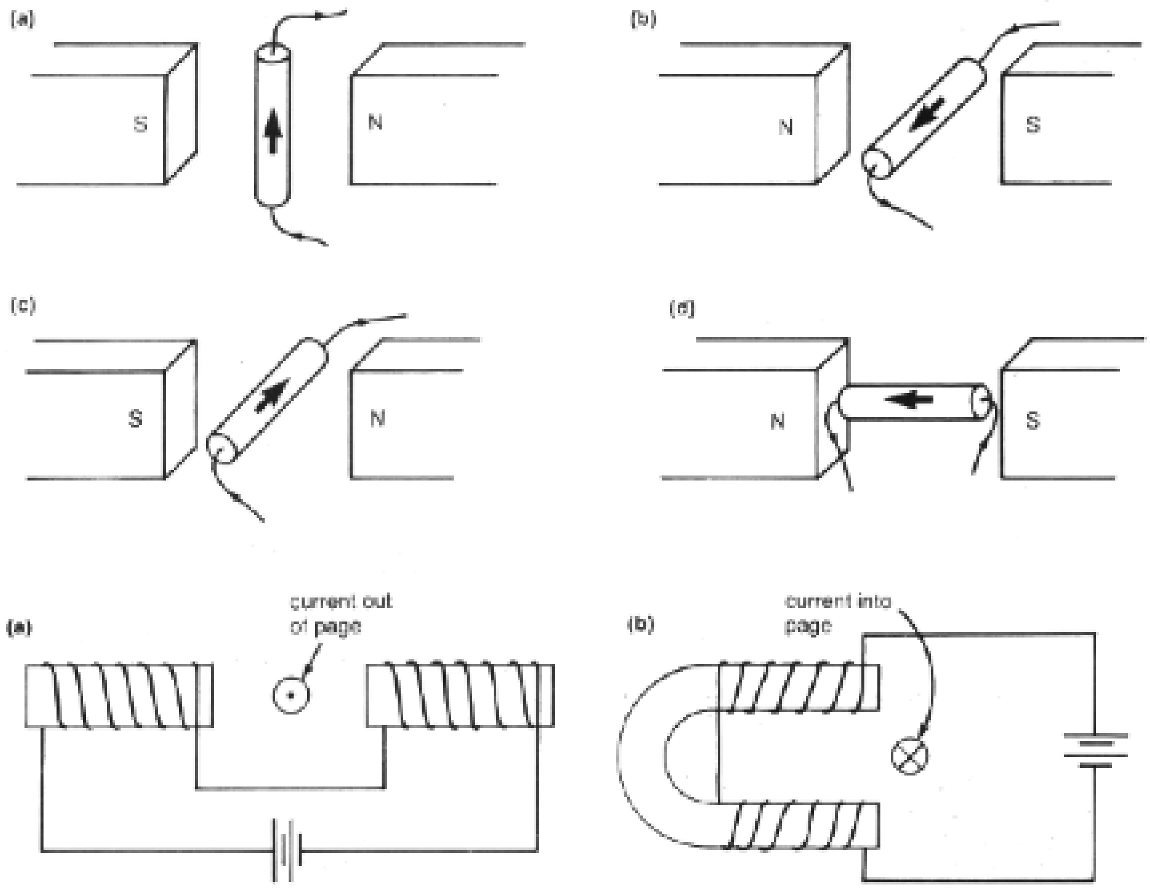
where F = force (N)
I = current flowing (A)
l = length of the conductor in the magnetic field (m)
B = magnetic field strength (T)

Note

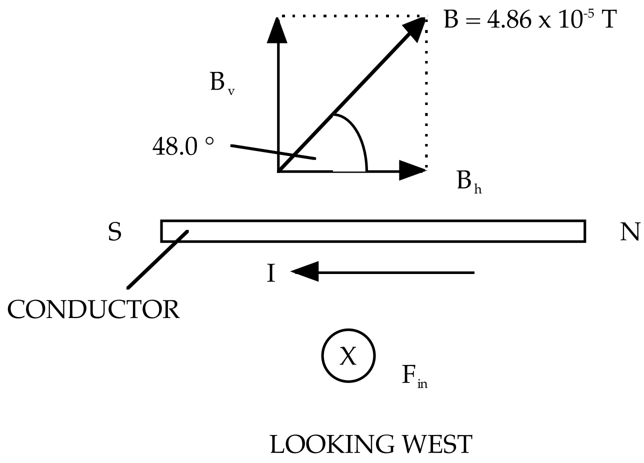
> The force is greatest when the conductor is at ***right angles to the magnetic field***.
>
> If the ***conductor is at an angle to the field***, then the ***component of B*** that is at right angles to the conductor is used.

Examples

1. On the following diagrams, mark the direction of the force acting on the conductor.



2. A straight conductor carrying a D.C. current of 1.50 A is arranged horizontally in a north-south direction in the southern hemisphere at a point where the Earth's magnetic field intensity is 4.86 x 10^{-5} T at 48.0° dip. Calculate the force exerted onto the conductor if the length of conductor in the field is 56.0 cm and the current flows north-south.



Using the R.H. palm rule, the force exerted is into the page (towards the west).

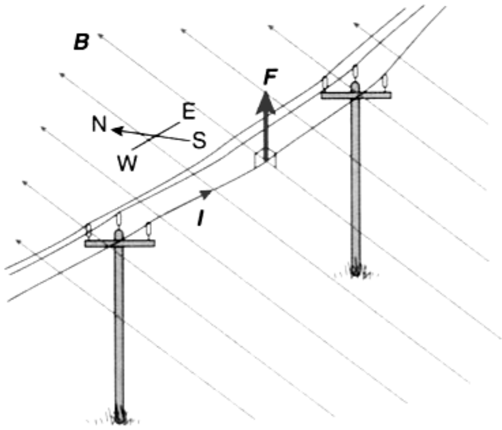
B_v is the component at right angles to the conductor.

$$
\begin{aligned}
F &= IlB \\
&= (1.50)(0.560)(4.86 \times 10^{-5} \cos 42.0°) \\
&= 3.034 \times 10^{-5} \text{ N}
\end{aligned}
$$

$\therefore$ Force = 3.03 x 10^{-5} N west.

3.

Determine the magnetic force per metre, due to the Earth's magnetic field, on a suspended power line running east–west and carrying a current of 100.0 A, as shown in the diagram. Assume the magnetic field is at a magnetic inclination of 55.9° up from horizontal.



Solution

As the line is running east–west, the current and magnetic field are at right angles. If we assume the Earth's magnetic flux density is 5.34×10^{-5} T, the magnitude of the force on 1.00 m of this power line is given by:

$\boldsymbol{B}_{\perp} = 5.34 \times 10^{-5}$ T $F_B = IlB_{\perp}$

$I = 100.0$ A $= (100.0)(1.00)(5.34 \times 10^{-5})$

$l = 1.00$ m $= 5.34 \times 10^{-3}$ N per metre at 90° to l

The direction of the force will be 55.9° up from horizontal to the south. In practice, the current in a power line is normally AC and so the force would alternate in direction from 55.9° up from horizontal to the south to 55.9° down from horizontal to the north.

4.

If the 100.0 A power line in Worked Example 3.2A runs north–south instead of east–west, and the magnetic inclination of the Earth's field is 55.9° up from horizontal, what would be the force on the line per metre of line?

Solution

This time the field and current are not at right angles and so we need to determine $\boldsymbol{B}_{\perp}$. The magnetic inclination is the angle of the field up from the horizontal and so it is also the angle between the current and the field.

$\boldsymbol{B}_{\perp} = 5.34 \times 10^{-5}$ T $\boldsymbol{B}_{\perp} = \boldsymbol{B} \times \sin\theta$

$\theta = 55.9°$ $= (5.34 \times 10^{-5})(\sin 55.9°)$

$= 4.42 \times 10^{-5}$ T

Now we can calculate the force on the wire

$\boldsymbol{B}_{\perp} = 4.42 \times 10^{-5}$ T $F_B = IlB_{\perp}$

$I = 100.0$ A $= (100.0)(1.00)(4.42 \times 10^{-5})$

$l = 1.00$ m $= 4.42 \times 10^{-3}$ N per metre at 90° to l

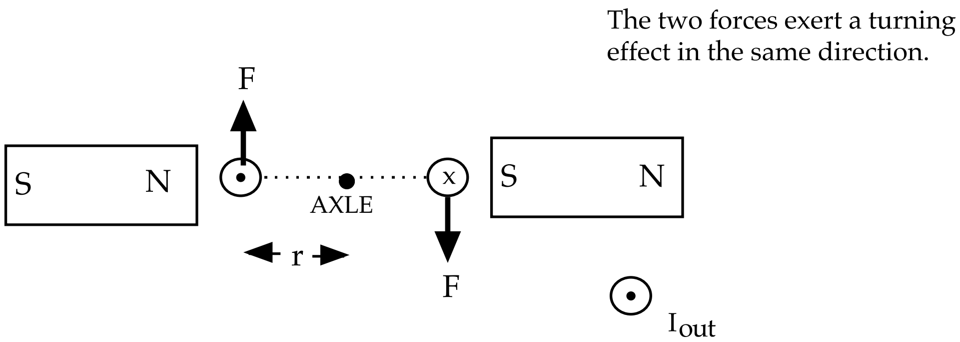
The right-hand rule enables us to find the direction of the force as east if the current is flowing north, or to the west if the current is flowing south.

ELECTRIC MOTOR

The force exerted by a magnetic field on a current-carrying conductor has many applications, the most important being the electric motor.

The basic principle of the motor is a rectangular loop of wire that is mounted on an axle between the poles of a magnet. The loop is usually pivoted along its long axis.

A force is exerted along each side of the loop that is at right angles to the magnetic field.



These forces exert a turning effect on the coil that is in the same direction. This arrangement is called a ***couple***.

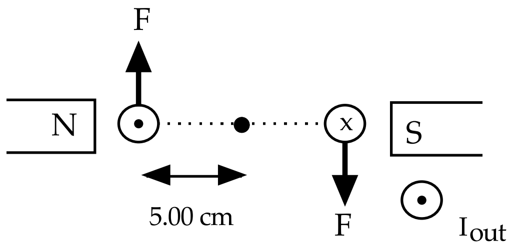
The magnitude of the torque generated by the coil is given by:

$$\boxed{\begin{aligned} \mathbf{M} &= \mathbf{2 \times N \times Fr} \\ &= \mathbf{2 \times N \times IlB \times r} \end{aligned}}$$

where
M = maximum torque or moment (Nm)
N = number of loops in the coil
I = the current flowing in the coil (A)
l = length of the coil at right angles to the magnetic field (m)
B = magnetic field intensity (T)
r = radius of turn of the coil (m)
= 1/2 x the length of side parallel to the magnetic field.

Note
The “2” in the formula is important as it shows the presence of the couple - two moments acting in the same direction.

Example 1
A D.C. motor has 200 turns of wire arranged into a rectangular coil of dimensions 20.0 cm x 10.0 cm. It is arranged with its long axis at right angles to a horizontal magnetic field of 0.200 T. Calculate the maximum torque generated by the coil when a current of 5.00 A flows through it.



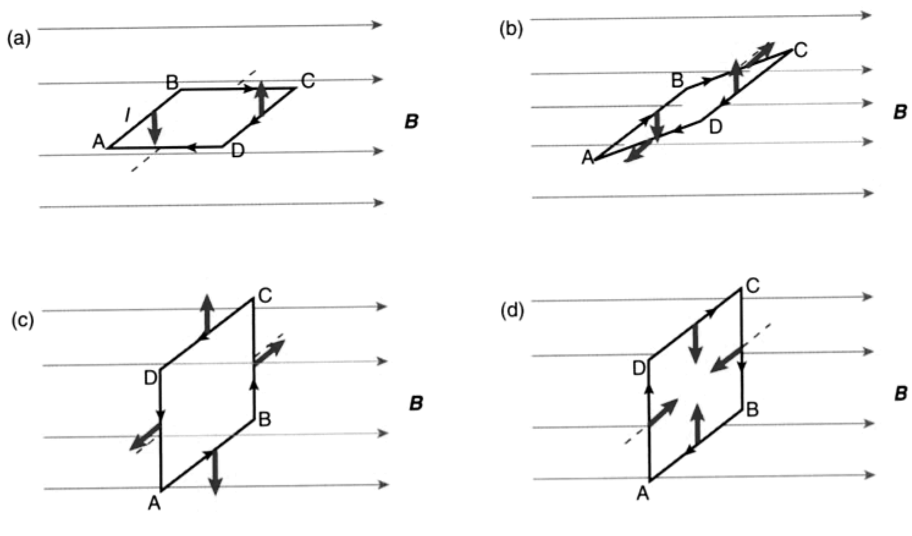
$$
\begin{aligned}
M &= 2 \times N \times F \times r \\
&= 2 \times N \times IlB \times r \\
&= 2\,(200)\,(5.00)\,(0.200)\,(0.200)\,(0.0500) \\
&= 4.00 \text{ Nm}
\end{aligned}
$$

$\therefore$ Maximum torque = 4.00 Nm clockwise .

Note
Each of the sides of the coil experiences a force due to the magnetic field. However, the magnitude varies as the coil rotates - it depends on the angle of each side to the magnetic field.

The forces on sides AB and CD create torque to turn the coil.

The forces on AD and BC only stretch the coil and have no effect on torque.



Example 2

The coil shown in Figure 3.28a is 4.00 cm square and consists of 20 turns of wire and carries a current of 5.00 A in a field of 0.800 T.

a What are the forces on each side?

b What is the torque on the coil?

Solution

a There are 20 turns in the coil so there are 20 lengths of wire on each side. The forces on sides AD and BC are zero because they are parallel to the field. The magnitude of the force on the other two sides is:

$N = 20$ turns

$I = 5.00$ A

$l = 0.0400$ m

$B_{\perp} = 0.800$ T

$$F = NIlB_{\perp}$$
$$= (20)(5.00)(0.0400)(0.800)$$
$$= 3.20 \text{ N}$$

The force on AB is down and on CD it is up.

b The torque on one arm of the coil is:

$F = 3.20$ N

$r_{\perp} = 0.0200$ m

$$\tau = Fr_{\perp}$$
$$= (3.20)(0.0200)$$
$$= 6.40 \times 10^{-2} \text{ N m}$$

The total torque will be twice this as both arms contribute the same torque, so:

$$\tau_{total} = 2 \times \tau$$
$$= 2(6.40 \times 10^{-2})$$
$$= 1.28 \times 10^{-1} \text{N m}$$

The direction of the torque is anticlockwise when viewed from the front.

Example 3

A loudspeaker is an application of the motor principle.

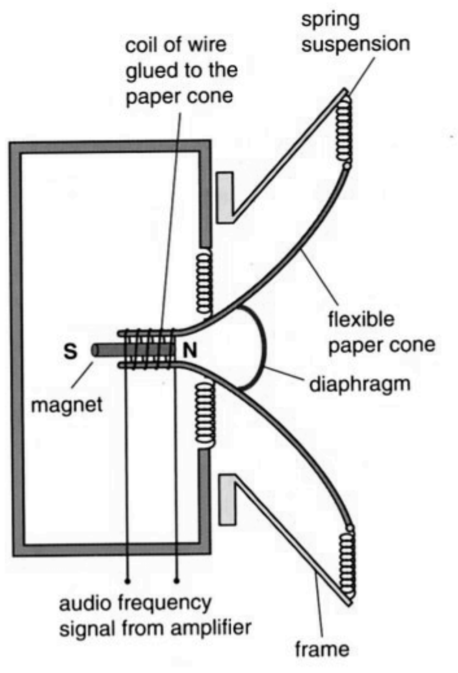
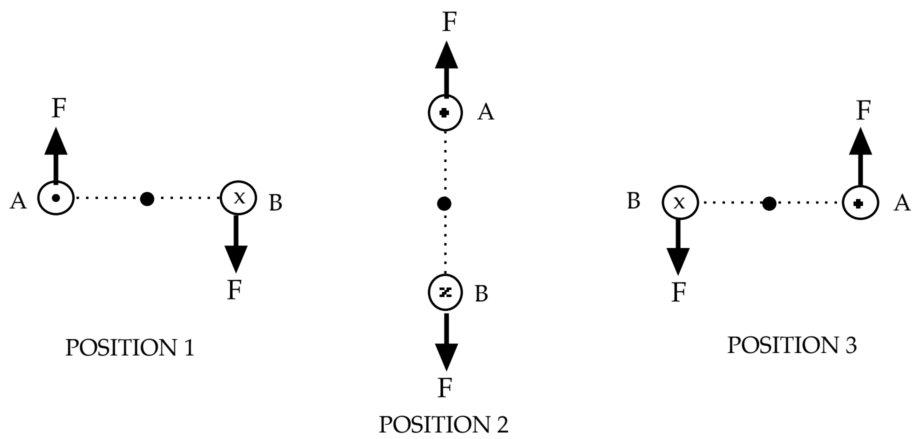


Figure 3.21

The principle on which a dynamic, or moving coil, loudspeaker operates. The changing current through the coil moves the magnet to and fro, making a paper or plastic cone vibrate, which in turn causes the air in front of it to vibrate. This can be clearly seen when the grill is removed from the front of a speaker, leaving it open to the air. A candle flame placed in front of the speaker will move back and forth.

DIRECT CURRENT (D.C.) MOTOR

Consider the simple motor shown previously and what happens to the torque produced as the coil turns.



POSITION 1: Maximum turning effect in a clockwise direction.

POSITION 2: ***Zero*** turning effect.

POSITION 3: Maximum turning effect but in the ***opposite direction***.

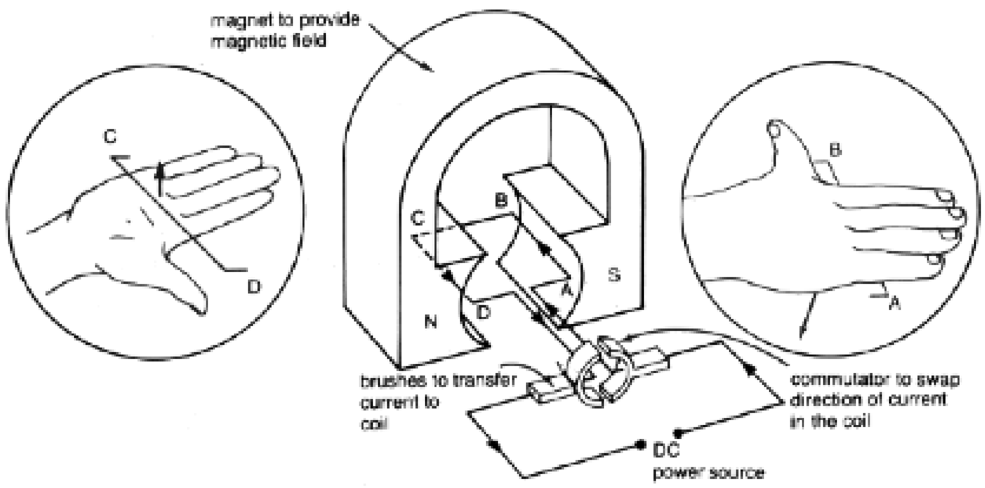
Commercial D.C. motors overcome the problem of a zero turning effect in position 2 by having ***several coils*** wound around a soft iron core (called an ***armature***) at angles to each other.

The soft iron increases the magnetic effect of the electric coils by concentrating their field.

By having the coils at an angle to each other, there will always be one coil in the correct position (parallel to the magnetic field) to experience the maximum turning effect. The motor will therefore turn with a constant torque value.

The problem of the coil reversing its direction every half-turn is overcome by using a ***split-ring commutator***. This will effectively reverse the current direction every half-turn so that the turning forces will always be turning the coil in the one direction.

The current is supplied to the commutator via small graphite or copper brushes.



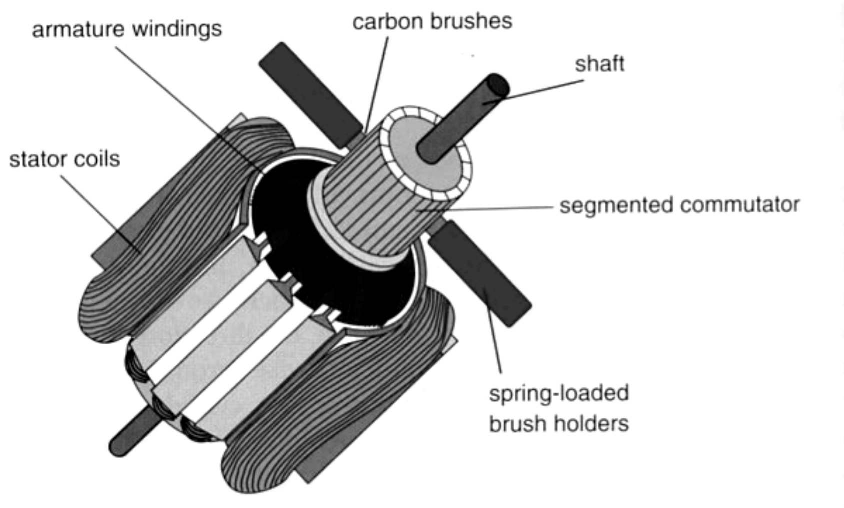


Figure 3.30
A typical universal electric motor, showing the main components. Some motors would have additional stator coils. The commutator feeds current to the armature coils in the position where most torque will be experienced. This type of motor will operate on either DC or AC current.

The power of the motor depends upon the total turning forces exerted on the coil.

These forces may be increased by:

- increasing the current flowing in the coil.
- increasing the strength of the magnetic field.
- increasing the number of loops in the coil.

ALTERNATING CURRENT (A.C.) MOTOR

This type of motor is called an ***induction motor***.

It does not need a split-ring commutator as the current is switching direction 50 times a second (50 Hz).

The construction of such a motor is complex and will not be examined in this course.

MOTION AND FORCES IN ELECTRIC AND MAGNETIC FIELDS

Electric Fields

A charged object has an electric field associated with it, and this influences other charged objects placed in it.

The electric field strength is defined as the force per unit charge.

i.e. $$\boxed{\mathbf{E} = \frac{\mathbf{F}}{\mathbf{q}}}$$

where
E = the electric field intensity (NC^{-1})
F = the force exerted on the charged object in the electric field (N)
q = the size of the charge on the object in the electric field (C)

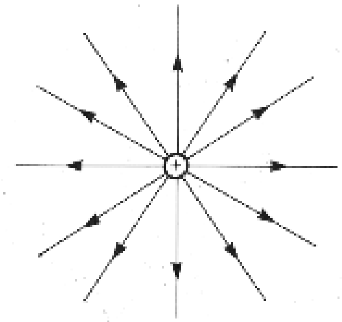
Electric field has both size (strength) and direction, hence it is a vector.

The direction of an electric field is taken as the direction a positive test charge, when placed in the field, would move.

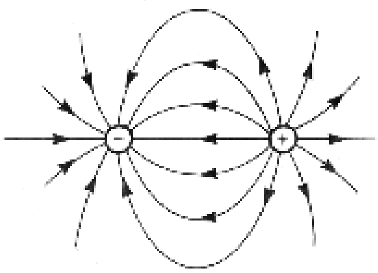
i.e. away from a positively charged object.

An electric field is represented by lines of force or flux lines.

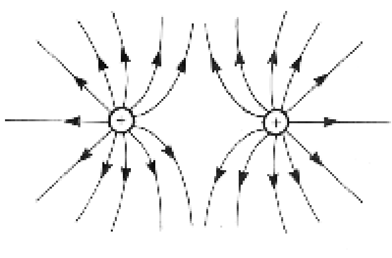
The electric field around various point and plate charges are as follows.



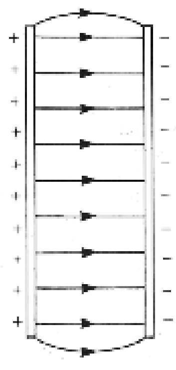
Isolated positive charge



Opposite point charges



Similar point charges



Parallel plates

Rules For Drawings

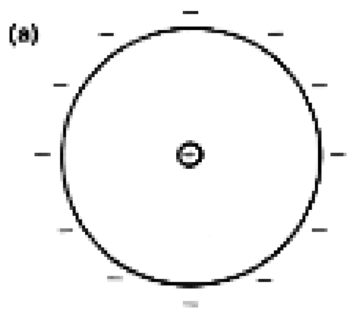
1. Electric field lines never touch nor cross each other.

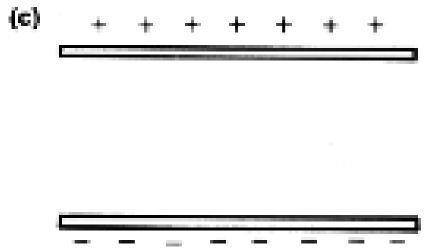
2. They enter and leave surfaces at right angles.

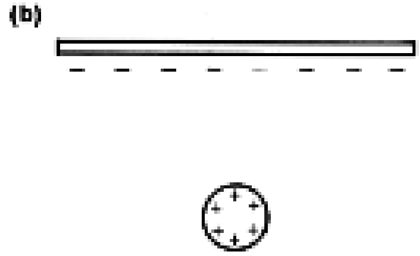
3. The intensity of the electric field is represented by the density or "bunching" of the field lines.

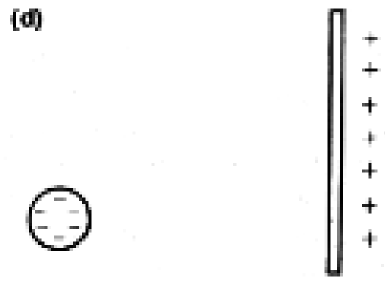
Examples

Draw the electric fields associated with the following.



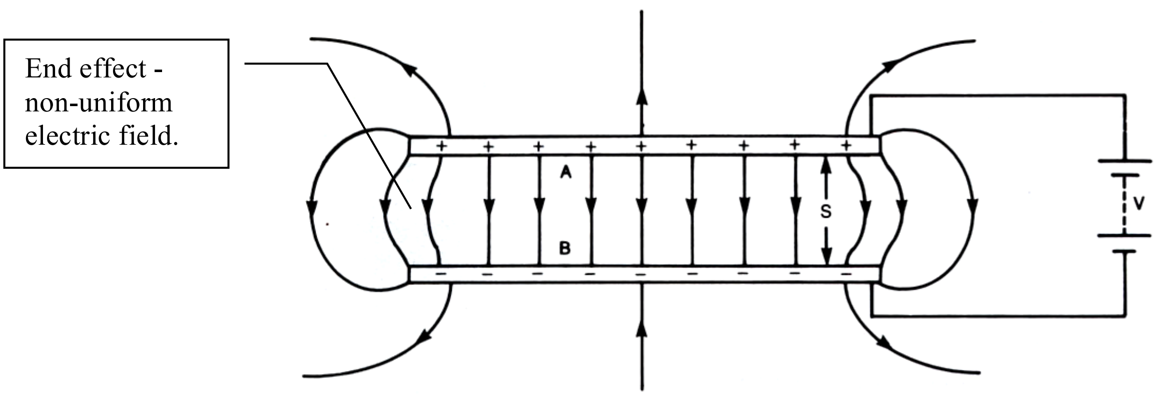






Electric Potential in Parallel Plates

The electric field lines between parallel plates are arranged as follows. For calculations, no "end effect" will be considered and the field is uniform throughout.



The electric field intensity between the plates is given by:

$$\boxed{\mathbf{E = \frac{V}{d}}}$$

where E = electric field intensity (Vm^{-1})
V = potential difference across the plates (V)
d = distance between the plates (m)

Note

> This relationship only applies to parallel plates.

Combining $E = \frac{F}{q}$ and $E = \frac{V}{d}$ gives:

$$\boxed{\mathbf{F = Eq = \frac{Vq}{d}}}$$

Potential difference is defined as the work done per unit charge in moving the charge through an electric field.

i.e. $$\boxed{\mathbf{V = \frac{W}{q}}}$$

where V = the difference in potential between two points in an electric field (V)
W = work done in moving the charge (J)
q = the amount of charge moved in the electric field (C)

When a charge is placed in an electric field, it experiences a force and moves. It moves from one potential to another, thus requiring some work to be done.

If work is done in moving the charge, it must change E_k.

i.e. $W = \Delta E_k$

=> $Vq = \frac{1}{2} mv^2 - \frac{1}{2}mu^2$

Since the charge starts from rest, $\frac{1}{2} mu^2 = 0$.

$\therefore$ $$\boxed{\mathbf{W = Vq = \frac{1}{2}mv^2}}$$

Examples

1. **Electron Gun** - used to accelerate electrons in older style TV's and computer monitors.

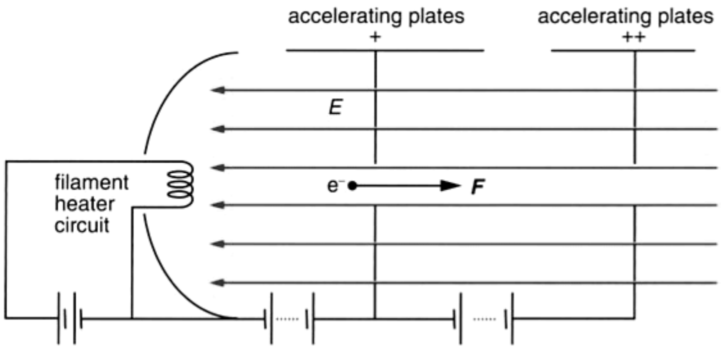


Figure 7.11

The set-up of a typical electron gun assembly. Electrons are released from a hot cathode and are accelerated across a potential difference through slits in a pair of positively charged plates. A magnetic field helps to centralise the path of the electrons.

An electron gun releases electrons from its cathode, which are accelerated across a potential difference of 32.0 kV, a distance of 30.0 cm between a pair of charged parallel plates. (Assume that the mass of an electron is 9.11×10^{-31} kg, the charge on an electron is -1.60×10^{-19} C and ignore any relativistic effects in your calculations.)

a Calculate the strength of the electric field acting on the electron beam.

b Calculate the magnitude of the velocity of the electrons as they exit the electron gun assembly.

c The electrons then travel through a uniform magnetic field perpendicular to their motion. Given that this field is of strength 0.200 T, calculate the expected radius of the path of the electron beam.

a $\Delta V = 32.0\times10^3$ V

$d = 30.0\times10^{-2}$ m

$$E = \frac{\Delta V}{d} = \frac{32.0\times10^3}{30.0\times10^{-2}} = 1.07\times10^5 \text{ V m}^{-1}$$

b $\Delta V = 32.0\times10^3$ V

$e = 1.60\times10^{-19}$ C

$m = 9.11\times10^{-31}$ kg

$$E_k = W_d$$

$$\tfrac{1}{2}mv^2 = e\Delta V$$

$$v = \sqrt{\frac{2e\Delta V}{m}} = \sqrt{\frac{2(1.60\times10^{-19})(32.0\times10^3)}{9.11\times10^{-31}}} = 1.06\times10^8 \text{ m s}^{-1}$$

This velocity is approximately 35% of the speed of light.

c $\boldsymbol{B}_\perp = 0.200$ T

$e = 1.60\times10^{-19}$ C

$m = 9.11\times10^{-31}$ kg

$v = 1.06\times10^8$ m s^{-1}

$$r = \frac{mv}{e\boldsymbol{B}_\perp} = \frac{(9.11\times10^{-31})(1.06\times10^8)}{(1.60\times10^{-19})(0.200)} = 3.02\times10^{-3} \text{ m}$$

This means the electrons will follow a path of radius 3.02 mm. In actual fact, this extremely small radius is not realistic, due to the effects of relativity.

2. Two parallel copper plates are 5.00 cm apart in a vacuum and have a potential difference of 1.00×10^3 V applied across them. Calculate:

(a) the electric field intensity between them.
(b) the force on an electron placed in the field.
(c) the velocity with which an electron will return to the positive plate if it is released from rest at the negative plate.

(a) $$E = \frac{V}{d} = \frac{1.00 \times 10^3}{5.00 \times 10^{-2}} = \underline{2.00 \times 10^4 \text{ Vm}^{-1}}$$

(b) $$F = Eq = (2.00 \times 10^4)(1.60 \times 10^{-19}) = \underline{3.20 \times 10^{-15} \text{ N}}$$

(c) $$W = Vq = {}^1/_2 mv^2$$

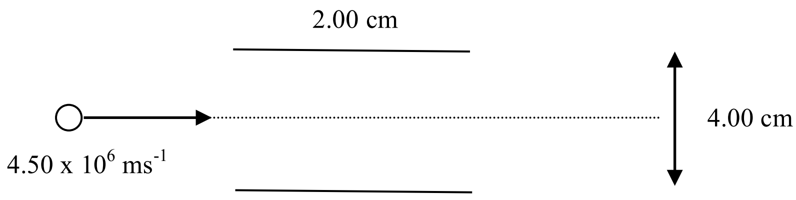
$$\Rightarrow \quad v = \sqrt{\frac{2Vq}{m}} = \sqrt{\frac{2(2.00 \times 10^4)(1.60 \times 10^{-19})}{9.11 \times 10^{-31}}}$$

$$= \sqrt{7.025 \times 10^{15}} = \underline{8.38 \times 10^7 \text{ ms}^{-1} \text{ towards the } +\text{ve plate}}$$

3. **Deflection Plates In Cathode Ray Tubes (CRT)**

A set of horizontal plates 2.00 cm long with a potential difference of 2.00×10^2 V across them is used to deflect electrons in an old TV set. An electron moving at $4.50 \times 10^6\ ms^{-1}$ is fired down the axis of the plates (in the middle). If the plates are 4.00 cm apart with the positive plate on top, determine the following.

(a) Calculate the acceleration of the electron due to the electric field.
(b) Should the effect of the acceleration due to gravity be considered?
(c) Calculate the vertical displacement of the electron while travelling between the plates.



(a) $$F_E = Eq = \frac{Vq}{d} = \frac{(2.00 \times 10^2)(1.60 \times 10^{-19})}{(4.00 \times 10^{-2})} = 8.00 \times 10^{-16}\ N$$

$$F = ma \quad \Rightarrow \quad a = \frac{F_E}{m} = \frac{8.00 \times 10^{-16}}{9.11 \times 10^{-31}} = \underline{8.78 \times 10^{14}\ ms^{-2} \text{ upwards}}$$

(b) The effect of gravity can be ignored. The acceleration of the force due to the electric field is 10^{14} times greater.

(c) The electron undergoes a similar path to a projectile moving horizontally under the effect of gravity.

Horizontally $$v_h = \frac{s_h}{t}$$

$$t = \frac{2.00 \times 10^{-2}}{4.50 \times 10^6} = 4.444 \times 10^{-9}\ s$$

Vertically

$v = ?$

$u = 0.0\ ms^{-1}$

$a = 8.78 \times 10^{14}\ ms^{-2}$

$t = 4.444 \times 10^{-9}\ s$

$s = ?$

$$s = ut + {}^1/_2 at^2$$
$$= 0 + {}^1/_2(8.78 \times 10^{14})(4.444 \times 10^{-9})^2$$
$$= 8.67 \times 10^{-3}\ m$$

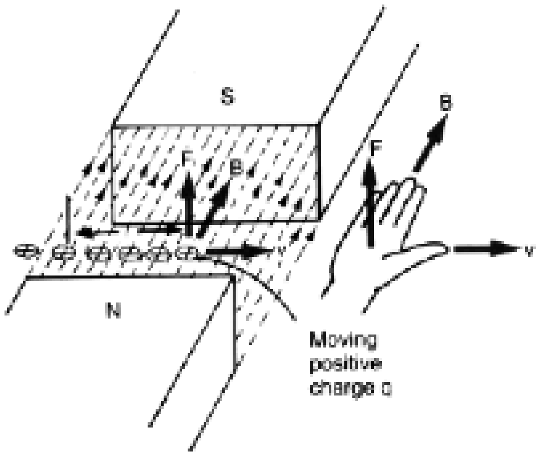
$\therefore$ <u>Electron moves 8.67 mm upwards</u>

Magnetic Fields

The force on a current-carrying wire is due to the individual force exerted on each of the moving charges within the conductor.

For an isolated charge moving in a magnetic field, it will be deflected as it is not constrained within a conductor.

The direction of the force is again determined by the ***right hand palm rule***. In this instance, the ***thumb is in the direction of the movement of a positive charge***.



If the moving charge is ***negative***, then the ***thumb is placed in the opposite direction*** to its movement.

The magnitude of the force is given by:

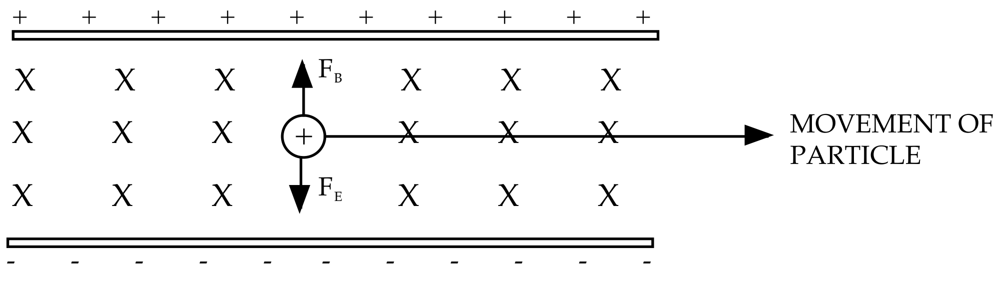
$$\boxed{\mathbf{F = qvB}}$$

where
F = force exerted at right angles to the path of the particle (N)
q = charge on the particle (C)
v = velocity of the particle (ms^{-1})
B = magnetic field intensity (T)

Note

> This equation will only apply when the velocity of the charges and the magnetic field are at ***right angles to each other***.

A method of measuring the velocity of a beam of charged particles is to pass the beam through a ***crossed electric and magnetic field*** as shown below.



Electric field is directed from the +ve to the -ve plate.
i.e. The direction a +ve charge would move.

The force due to the electric field is balanced by the force due to the magnetic field and the particle passes through undeflected.

i.e. $\boxed{\mathbf{F_E = F_B}}$

Only one velocity for the particles will allow them to pass through undeflected - hence it is called a ***velocity selector***.

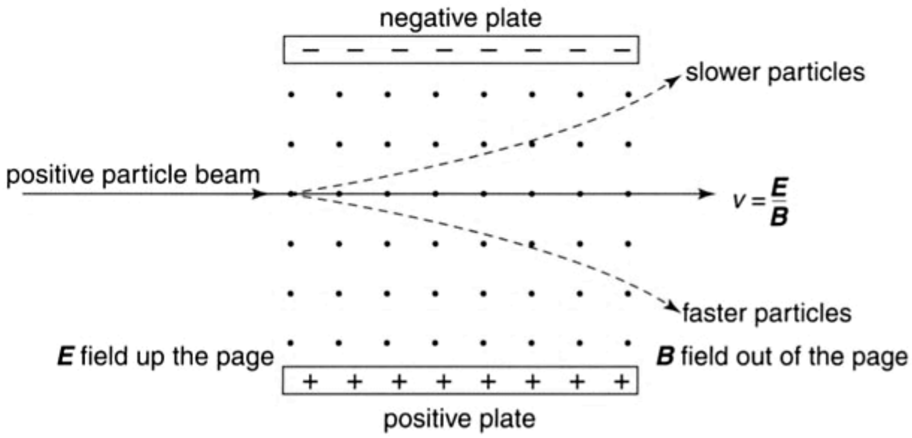


Figure 7.32
The crossed fields arrangement for a velocity selector. Only charged particles of one particular velocity will pass through the fields in a straight line. Faster or slower particles will travel in different paths because the size of the magnetic force depends on the speed of the particles, whereas the electric force is independent of the speed of the particles.

Examples

1. A beam of positrons passes undeflected through a pair of crossed electric and magnetic fields in which the electric field intensity is 3.70 x 10^3 Vm^{-1} and the magnetic field intensity is 7.86 x 10^{-2} T. What must be the velocity of the positrons?

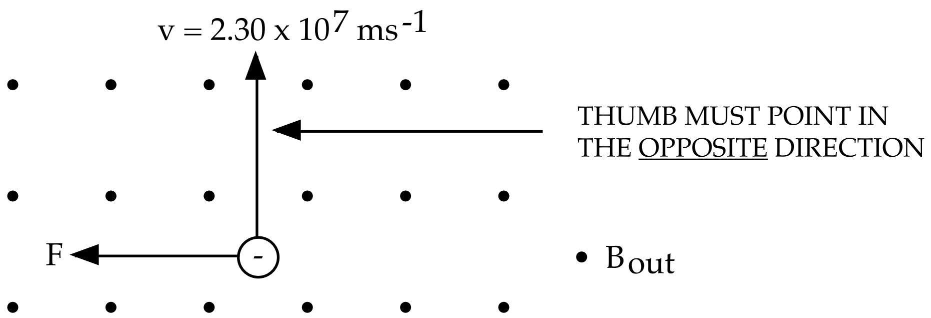
$$F_E = F_B$$
$$\Rightarrow \quad Eq = qvB$$
$$\Rightarrow \quad v = \frac{E}{B}$$
$$= \frac{3.70 \times 10^3}{7.86 \times 10^{-2}}$$
$$= 4.707 \times 10^4 \text{ ms}^{-1}$$

$\therefore$ The speed of the electrons = 4.71 x 10^{4} ms^{-1}

Note

The velocity of the particles is independent of the size of the charge on them.

2. A beam of negatively charged beta particles is directed south-north in a magnetic field of 3.56 x 10^{-2} T directed vertically upwards. If the velocity of the particles is 2.30 x 10^7 ms^{-1}, calculate the magnitude and direction of the force acting on each.



$$F = qvB$$
$$= (1.60 \times 10^{-19})(2.30 \times 10^{7})(3.56 \times 10^{-2})$$
$$= 1.310 \times 10^{-13} \text{ N}$$

$\therefore$ The force exerted = 1.31 x 10^{-13} N West.

SHAPE OF THE PATH OF A CHARGED PARTICLE IN A MAGNETIC FIELD

The force exerted on a charged particle is ***always at right angles to its direction of travel***. Thus, it does not alter the speed of the particle.

It will change its path, forcing the particle to move in a ***circular path***.

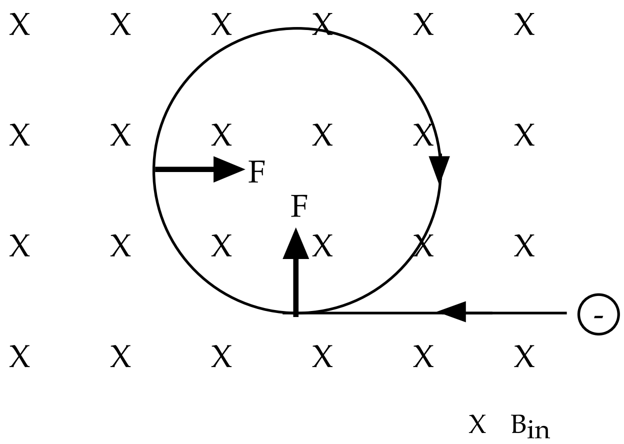
Thus, the force that acts is a ***centripetal force***.

$$\text{i.e.} \qquad F_c = F_{\text{due to B}}$$

$$\Rightarrow \qquad mv^2/_r = qvB$$

$$\Rightarrow \qquad \boxed{r = \frac{mv}{qB}}$$

where
r = radius of orbit of the charged particle (m)
m = mass of the particle (kg)
v = orbital speed of the particle (ms^{-1})
q = charge on the particle (C)
B = magnetic field intensity (T)



No matter where the "right hand palm rule" is applied along the path, the force will always be directed towards the centre of the curve.

Example

An electron gun operating at 1.35×10^3 V is used to accelerate electrons and fire them horizontally into an evacuated chamber where a vertical magnetic field of 0.114 T is directed upwards.

(a) What is the velocity of the electrons as they enter the magnetic field?
(b) Do the electrons move clockwise or anticlockwise?
(c) What is the period of orbit of the electrons?

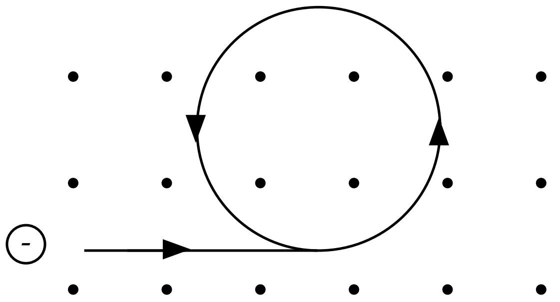
(a) The electrons in the gun are accelerated by an electric field that exists due to a potential difference between an anode (source of electrons) and a cathode.

$$\text{Electrical work} = E_k \text{ gained by the electrons}$$
$$\text{i.e.} \quad W = Vq = 1/2\, mv^2$$
$$\Rightarrow \quad (1.35 \times 10^{3})(1.60 \times 10^{-19}) = 1/2\,(9.11 \times 10^{-31})v^2$$
$$\Rightarrow \quad v = 2.178 \times 10^{7}\ \text{ms}^{-1}$$

$\therefore$ The velocity of the electrons = 2.18×10^{7} ms^{-1} horizontally.

THIS VALUE CAN NEVER BE GREATER THAN 3.00×10^8 ms^{-1}. MAKE SURE YOU TAKE THE SQUARE ROOT!

(b)



Electrons travel **anticlockwise**.

• B$_{out}$

(c)

$$r = \frac{mv}{qB}$$
$$= \frac{(9.11 \times 10^{-31})(2.178 \times 10^{7})}{(1.60 \times 10^{-19})(0.114)}$$
$$= 1.088 \times 10^{-3}\ \text{m}$$

$$v = \frac{2\pi r}{T}$$
$$\Rightarrow \quad T = \frac{2\pi r}{v}$$
$$= \frac{2\pi(1.088 \times 10^{-3})}{(2.178 \times 10^{7})}$$
$$= 3.139 \times 10^{-10}\ \text{s}$$

$\therefore$ Period of revolution = 3.14×10^{-10} s.

Examples
Below are some common examples and applications of electric charges moving in a magnetic field.

1. **Television, Cathode Ray Oscilloscopes (C.R.O.)**

   Electrons are made to "sweep" the screen using electric coils producing magnetic fields to deflect electrons vertically and horizontally.

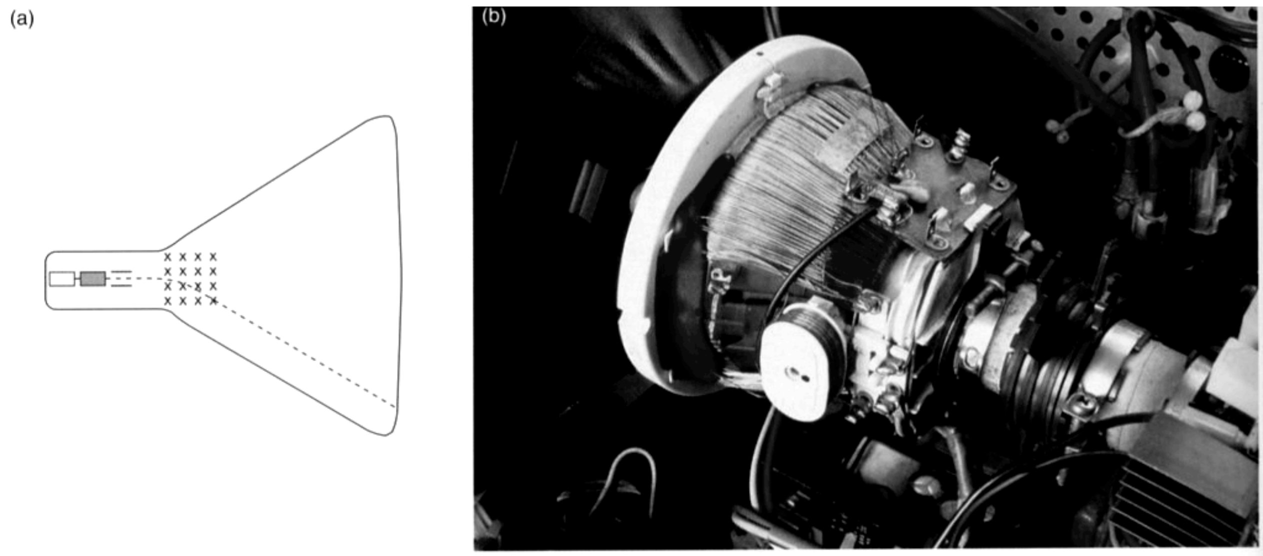


Figure 7.1
The deflection coils, or yoke, at the neck of a computer monitor cathode ray tube cause the beam to be deflected through a large angle, so the screen can be made reasonably short. The current in the coils varies rapidly in order to create the magnetic field which causes the beam to sweep across the screen at the same time as it is moving down the screen. Altogether it traces out 625 horizontal lines in two vertical sweeps of the screen, 25 times every second.

2. **Mass Spectrometer**

   Ions and other charged particles are accelerated by electric fields and fired into a magnetic field. The particles travel in a circular path until they collide with a detector (photographic film in the early machines, a computer-controlled detector in recent machines).

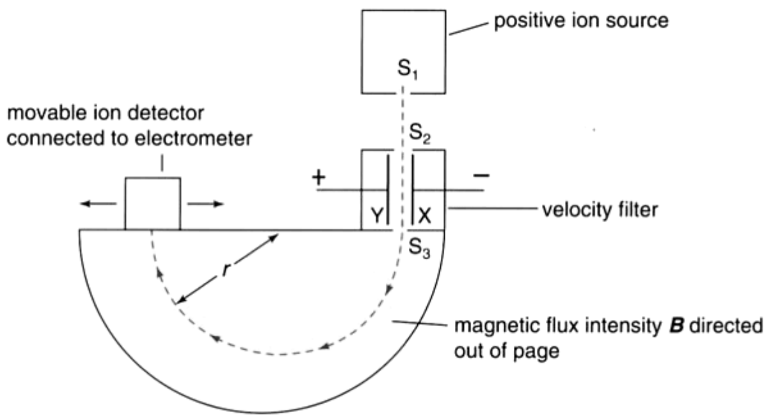


Figure 7.31
The Bainbridge spectrometer. Ions of uniform velocity enter a magnetic field at 90° to the field and continue in a circular path.

A beam of singly charged nitrogen molecules is fired into a mass spectrometer after being filtered through crossed fields of 4550 V m^{-1} and 0.120 T. What would be the radius of the semicircular path that they would follow in the spectrometer? The mass of nitrogen molecules is 4.68×10^{-26} kg.

Solution

First, find the velocity of the molecules:

$\boldsymbol{E} = 4550\ \text{V m}^{-1}$

$\boldsymbol{B}_\perp = 0.120\ \text{T}$

$$v = \frac{\boldsymbol{E}}{\boldsymbol{B}_\perp} = \frac{4550}{0.120} = 3.79 \times 10^4\ \text{m s}^{-1}$$

Now we can calculate the radius. Note that a singular charge is 1.60×10^{-19} C:

$m_N = 4.68 \times 10^{-26}$ kg

$\boldsymbol{B}_\perp = 0.120$ T

$v = 3.79 \times 10^4\ \text{m s}^{-1}$

$q = 1.60 \times 10^{-19}$ C

$$r = \frac{mv}{q\boldsymbol{B}_\perp} = \frac{(4.68 \times 10^{-26})(3.79 \times 10^4)}{(1.60 \times 10^{-19})(0.120)} = 9.24 \times 10^{-2}\ \text{m}$$

Hence, the radius is equal to 9.24 cm.

Physics in action — Cyclotrons

To perform experiments at very high energies, linear accelerators would need to be extremely long. For this reason, the American physicist Ernest O. Lawrence designed the first circular accelerator in the 1930s. This is called a *cyclotron*, and it won Lawrence a Nobel Prize in 1939. In some respects, the cyclotron operates as a spiral-shaped linac. Protons are often used as the accelerating particles in this machine.

Here, the many drift tubes are replaced by two semicircular, D-shaped, hollow copper chambers, called dees. These are the positive and negative electrodes of the cyclotron between which exists a strong electric field. The dees sit back to back, giving the cyclotron its circular shape, and lie between the poles of a powerful electromagnet. The inside of the metallic dee is shielded from the electric field. The magnetic field acts on the particles, producing a circular path. When a particle emerges from the dee, the sign of the accelerating potential is reversed, so the particle speeds up towards the other dee. This occurs so that a proton will accelerate towards a negatively charged dee as it exits the positively charged dee. Each time the particles cross the gap between the dees, their speed increases and they travel in a semicircle of larger radius. They gain energy with each revolution until they attain sufficient energy to exit the accelerator.

A key to the operation of the cyclotron is that the frequency of the radio-frequency generator (RF generator) that produces the alternating field must match the frequency of the circulating charged particles. The charged particles travel in a path of radius:

$$r = \frac{mv}{q\boldsymbol{B}_\perp}$$

Their speed is then:

$$v = \frac{rq\boldsymbol{B}_\perp}{m}$$

and the time taken for one orbit of the cyclotron is:

$$T = \frac{d}{v} = \frac{2\pi r}{v}$$

Substituting for r in the above expression:

$$T = \frac{2\pi r}{v} = \frac{2\pi mv}{vq\boldsymbol{B}_\perp} = \frac{2\pi m}{q\boldsymbol{B}_\perp}$$

Strangely enough, the time taken for one revolution of the cyclotron does not depend upon the velocity of circulating charges. This is because as the speed increases, the radius of path travelled also increases and the time taken for each orbit remains the same.

In 1943, the Adelaide-born physicist Marcus Oliphant, while working in Britain, suggested modifying the cyclotron design to produce a synchrotron.

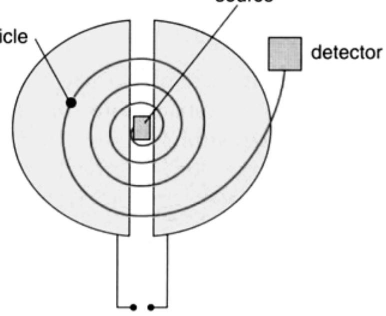


Figure 7.16

The cyclotron operates in some ways like a spiral linac—particles are accelerated from a source, through semicircular chambers called dees, until they gain sufficient energy to exit.

Physics in action — Mass spectrometers and particle accelerators

One of the most powerful tools of the modern chemist is the mass spectrometer. It enables the very accurate determination of the mass of the atoms in a tiny sample of material. This can help chemists to identify the substances in very small samples, for example tiny amounts of a rare element in a sample of Moon rock, or the mud on the shoes of a suspected criminal.

The sample is heated in an evacuated container until it is ionised, and the positive ions are accelerated electrically to high speeds. They then enter a magnetic field where they experience a '***qvB***' force which will cause them to move in a circle. The radius of this circle depends on the mass of the particle: heavier particles with the same charge and speed will experience the same force, but because of their greater inertia they will move in a larger circle. Thus ions of different masses end up in a different place on the detector. As the relative amounts of each ion can also be measured, the composition of the original sample can be determined. (A new type of mass spectrometer, called an *ion trap*, does not use this principle at all. It uses high-frequency radio waves to trap ions of a certain mass.)

To study the basic constituents of matter, physicists accelerate particles (such as electrons or protons) to very high speeds and then let them crash into atoms. The results of these collisions have revealed that an incredible array of subatomic particles exists. Understanding the properties of these particles is fundamental to understanding our Universe.

The electrons or protons are accelerated by electric fields of various sorts, but very long paths are needed to obtain the extremely high speeds necessary (close to the speed of light). Because it is impractical to have a straight path hundreds of kilometres long, the particles travel through very strong magnets which cause them to move in a circle. The circular accelerator at CERN in Switzerland is nearly 9 km in diameter! Closer to home, the new synchrotron near Monash University in Melbourne is 70 m in diameter. It can be much smaller because it accelerates electrons, not protons, and it uses more powerful magnets to bend the electrons into the circular path.

The Australian Synchrotron accelerates electrons to a massive 3 GeV of energy. This is equivalent to accelerating them through 3000 million volts. At this energy, they travel at about 99.99999% of the speed of light. Because of relativistic effects, their effective mass at that speed is about 6000 times the rest mass. The path of the electrons is not so much a circle, but a series of straight sections in between 24 bending magnets 2 m long and with field strengths of about 1.5 T. As the electrons go through these magnets, because they are being accelerated, they give off energy in the form of electromagnetic radiation—light. As well as the bending magnets, there are other magnets called undulators and wigglers which, as the names imply, accelerate the electrons back and forth, again causing them to give off enormously intense light. It is this light, ranging from infrared and visible to hard X-rays, that is used for many different research purposes.

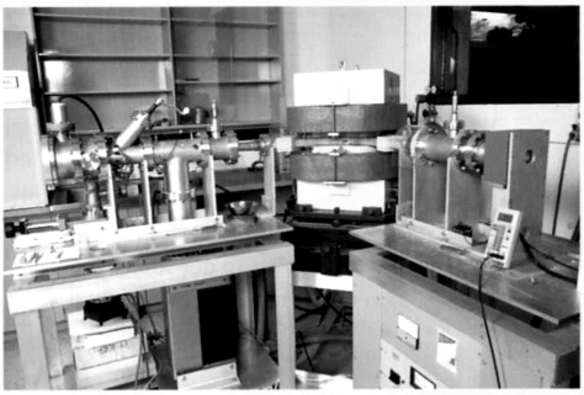


Figure 7.6
A simple mass spectrometer

Figure 7.7
The Australian Synchrotron